

- Statistical indicators

Creating a float variable:

```
1. stock_price = 142.58
2. interest_rate = 4.25
3. percent_change = -2.35
```

?? Fun Fact: During the Knight Capital disaster of 2012, the firm lost \$440 million in just 45 minutes due to a trading algorithm gone wrong. One of the issues involved imprecise calculation of decimal values in their automated system. The catastrophic error, which nearly bankrupted the company, could have been prevented with proper handling of floating-point numbers and thorough testing—a stark reminder of how important proper data type management is in financial algorithms.

Strings: Text Information

Strings (or `str`) store text information. In trading algorithms, they're used for:

- Stock symbols
- Company names
- Market status messages
- Data sources

Creating string variables:

```
1. stock_symbol = "AAPL"
2. company_name = "Apple Inc."
3. market_status = "Open"
```

Note that strings are always enclosed in quotes (either single `'` or double `"`).